

HIL Sensor Attachment & Soldering Guide

QT Py ESP32-S3 N4R2 - TCA9548A mux @ 0x77 - generated from hil_map.json

1. Soldering / jumper adjustments (9 boards)

These boards move off their default address to coexist. All others sit at their factory/default address - no soldering.

Sensor	Driver	Channel	Default	Target	Jumper
HTU31D	drvHtu31d	TCA9548A	0x40	0x41	ADDR:VDD
BMP581	drvBmp5xx	TCA9548A	0x47	0x46	SDO:GND
BME280	drvBme280	TCA9548A	0x77	0x76	Solder ADDR jumper on back (or wire SDO to GND). D
HDC302x	drvHdc302x	TCA9548A	0x44	0x45	A0:1
INA228	drvIna228	TCA9548A	0x40	0x41	A0:1
INA237	drvIna237	TCA9548A	0x40	0x45	A0:1 A1:1
INA238	drvIna238	TCA9548A	0x40	0x41	A0:1
INA260	drvIna260	TCA9548A	0x40	0x45	A0:1 A1:1
SPA06-003	drvSpa06_003	TCA9548A	0x77	0x76	SDO:GND

2. Attachment list - PR #933 drivers + currently attached (32)

Grouped by bus/mux channel. Green = already attached. Amber = needs soldering. Find and attach the NEED rows.

Direct Bus (No Mux)

Sensor	Driver	Addr	Jumper / note	Status
D6T-1A Thermal Sensor	drvD6t1a	0x0A	default/fixed	NEED
PMSA003I	drvPm25	0x12	On the string; not one of the 2	ATTACHED
VCNL4200	drvVncl4200	0x51	default/fixed	NEED
SGP30	drvSgp30	0x58	default/fixed	ATTACHED
SEN66	drvSen6x	0x6B	default/fixed	NEED

TCA9548A (0x77) Ch0

Sensor	Driver	Addr	Jumper / note	Status
QMC5883P	drvQmc5883p	0x2C	default/fixed	NEED
AS5600 Magnetic Angle	drvAs5600	0x36	default/fixed	NEED
MLX90632-B (Standard)	drvMlx90632	0x3A	default/fixed	NEED
MLX90632 (Extended Mode)	drvMlx90632	0x3B	default/fixed	NEED
HTU31D	drvHtu31d	0x41	set jumper 'ADDR:VDD' (default	NEED
BMP581	drvBmp5xx	0x46	set jumper 'SDO:GND' (default	NEED
BMP580	drvBmp5xx	0x47	default/fixed	NEED
MAX44009	drvMax44009	0x4A	default/fixed	NEED
APDS9999	drvApds9999	0x52	On the string; not a PR #933 d	ATTACHED
SGP40	drvSgp40	0x59	On the string now. Shares 0x5	ATTACHED
SCD30	drvScd30	0x61	Reattached; not a PR #933 dri	ATTACHED
STCC4	drvStcc4	0x64	default/fixed	NEED
AS7331	drvAs7331	0x74	default/fixed	ATTACHED
BME280	drvBme280	0x76	set jumper 'Solder ADDR jump	ATTACHED

TCA9548A (0x77) Ch1

Sensor	Driver	Addr	Jumper / note	Status
MLX90632	drvMLX90632	0x3A	default/fixed	NEED
HDC302x	drvHdc302x	0x45	set jumper 'A0:1' (default 0x44	NEED
BMP585	drvBmp5xx	0x47	default/fixed	NEED
TMP119	drvTmp119	0x48	default/fixed	ATTACHED
ENS160	drvEns160	0x52	On the string; not a PR #933 d	ATTACHED
SGP41	drvSgp41	0x59	Needs a REAL SGP41 - the attached SGP40 Part	NEED
LPS28DFW	drvLps28dfw	0x5C	default/fixed	NEED
VCNL4030	drvVcnl4030	0x60	default/fixed	NEED

TCA9548A (0x77) Ch2

Sensor	Driver	Addr	Jumper / note	Status
INA228	drvIna228	0x41	set jumper 'A0:1' (default 0x40	NEED
INA237	drvIna237	0x45	set jumper 'A0:1 A1:1' (default	NEED

TCA9548A (0x77) Ch3

Sensor	Driver	Addr	Jumper / note	Status
INA238	drvIna238	0x41	set jumper 'A0:1' (default 0x40	NEED
INA260	drvIna260	0x45	set jumper 'A0:1 A1:1' (default	NEED

TCA9548A (0x77) Ch7

Sensor	Driver	Addr	Jumper / note	Status
SPA06-003	drvSpa06_003	0x76	set jumper 'SDO:GND' (default	NEED